

Does 3:1 Ever Look Like 3:1?

What a Mendelian ratio actually predicts about one batch of offspring, and what chi-square is really asking when you test it.

Name _____

Date _____

Period _____

Demonstration: andresdbp.github.io/resources/demos/does-3-1-look-like-3-1/

1 Before you open anything

Answer from what you already know about monohybrid crosses. Nothing here is graded on being right — the point is to have a number on paper that the demonstration can later confirm or contradict.

PREDICTION

A cross is *exactly* 3:1: every offspring has probability 0.75 of the dominant phenotype, with no linkage, no lethal alleles and no scoring error. You count **20** offspring, then repeat that whole count many times over.

Write the *lowest* and the *highest* dominant : recessive ratio you expect to see across those repeats.

Now suppose you counted **1000** offspring in each repeat instead. Would that range of ratios be wider, narrower, or the same? Circle one, then say in one sentence how you decided.

2 Reading one cross

Open the demonstration. The top panel, *The experiment*, draws one batch of offspring; the panel below it, *Chi-square, term by term*, tests that batch against 3:1.

1. Press **Reset**. The line under the buttons should read **Run 1** and *Offspring counted, n* should read **40**.
2. Copy the chi-square table exactly as the demonstration shows it, one row per phenotype class.
3. Take the dominant row and compute $(O - E)^2/E$ yourself in the last column of your own table. Confirm you get the number on screen.

Class	O	E	O - E	(O-E) ² /E
Dominant				
Recessive				
Total				

a. Record the four tiles under that table: χ^2 = _____ p-value = _____ Degrees of freedom = _____ Critical value, $\alpha = 0.05$ = _____

b. There are two phenotype classes, yet the degrees of freedom is 1. Read the hint printed under the table and state the reason in your own words.

1. Now click **20** in the *Offspring counted, n* row. Record the first line of the table below.
2. Press **Re-run the cross** and record the next line. Repeat until all six lines are filled. Do not change anything else.

Run	Observed dominant	Observed recessive	Observed ratio	χ^2	Reject or fail to reject?
1					
2					
3					
4					
5					
6					

c. Across those six runs, how much did the probability of a dominant offspring change? Quote the hint under the *Offspring counted, n* buttons to justify your answer.

Leave n at 20 and read the orange panel that has appeared under the verdict. The expected recessive count is 5.0, sitting exactly on the rule-of-thumb floor of about 5 expected per class, and with 20 offspring the observed percentage can only land on multiples of 5. Treat every p-value you record at n = 20 as indicative rather than exact.

3 What sample size buys you

One control, five settings. The underlying cross is exactly 3:1 at every one of them, so anything that changes in this table is caused by sample size alone.

1. Press **Reset** again so you start from Run 1 with *Replicate experiments* at **300**.
2. Click each value of *Offspring counted, n* in turn and read the tiles in the *Run the whole experiment again and again* panel. Do not press Re-run between settings.
3. Fill the last column yourself: multiply the predicted spread by $\sqrt{n}$. Round to the nearest whole number.

n	Spread of results, observed (pp)	Spread predicted by theory (pp)	Range of observed ratios across the replicates	Predicted spread $\times \sqrt{n}$
20				
40				
100				
400				
1000				

a. The last column barely moves. Use that to complete the rule: the standard deviation of the percentage dominant is roughly _____ $\div \sqrt{n}$ percentage points.

b. Going from $n = 20$ to $n = 1000$ multiplies the sample size by 50. By what factor did the spread shrink? Show how that factor follows from your rule in part a.

c. Compare the $n = 20$ and $n = 1000$ rows of the *range of observed ratios* column with the two answers you committed to on page 1. Say specifically what your reasoning assumed that the data did not support — or, if you were right, what evidence in this table confirms it.

d. The horizontal axis of the histogram is fixed from 40% to 100% at every setting of n. Explain why holding that axis still is essential to the comparison you just made.

4 Where the eyeball test fails

A common shortcut is to judge a cross by how far its ratio *looks* from 3.00:1. Below, that shortcut is put up against the test itself.

1. Click **20**, then **100**, then **1000** in the *Offspring counted, n* row. At each one, read the note at the bottom of the *Run the whole experiment again and again* panel: it states the range of dominant counts a chi-square test would accept at that n.
2. Convert each end of that count range into a dominant : recessive ratio.

n	Accepted counts, dominant	Lowest accepted ratio	Highest accepted ratio
20			
100			
1000			

a. A cross gives **72% dominant offspring**. Write that as counts at n = 100 and again at n = 1000, then compute χ^2 by hand for each. Use $E = 0.75n$ and $E = 0.25n$.

b. State the verdict at each sample size against the critical value 3.841, and check both against the accepted-count ranges in your table.

c. The ratio was identical in both cases. Explain why the two verdicts differ, and why that means “this ratio looks close enough to 3:1” is not a test.

Look at the two rejection tiles while you work. About 5% of the replicates get rejected at $\alpha = 0.05$ even though every one of them came from a cross that is exactly 3:1. The tile marked *Exact long-run rate at this n* gives the true figure for the n you have selected; it is never a clean 5.00%, because offspring come in whole numbers and the cut can only fall between two adjacent counts.

5 Solve these without the demonstration

Close the page or look away. Show every step: expected counts, each $(O - E)^2/E$ term, the total, the degrees of freedom, and the comparison with 3.841. Then reopen the demonstration, set n to match, and check your verdict against the accepted-count range in the third panel.

1. A monohybrid cross yields 40 offspring: 28 with the dominant phenotype and 12 with the recessive. Test the 3:1 hypothesis at $\alpha = 0.05$ and state your conclusion in a full sentence.

2. The same cross is repeated on a larger scale: 400 offspring, 285 dominant and 115 recessive. Test the 3:1 hypothesis at $\alpha = 0.05$.

3. A third count gives 1000 offspring: 712 dominant and 288 recessive. Test the 3:1 hypothesis at $\alpha = 0.05$.

4. No calculation for this one. Problems 2 and 3 both gave about 71% dominant, but only one of them was rejected. A lab partner concludes that the 1000-offspring count must have been done badly. Explain what is wrong with that conclusion, and give two things a significant χ^2 does *not* tell you.

6 On your own

Nothing here is graded, and the demonstration does more than this worksheet used.

- Drag the *Replicate experiments* slider from 20 up to 500 and watch the two rejection tiles. One of them jumps around and one of them does not. Work out which is the estimate and which is the target, and how many replicates you need before they agree to within a percentage point.
- Step through all five values of n reading only the *Exact long-run rate at this n* tile. It is never exactly 5.00%. Find which n comes closest, and explain the pattern using the fact that offspring counts are whole numbers.
- At $n = 20$ the histogram is a comb of evenly spaced bars with gaps between them. Measure the spacing, account for it, and then find where the replicates that fall below 40% dominant end up on that axis.